

ASWIN M

Chennai, India | aswinmurugan2712@gmail.com | 9342798338 | www.linkedin.com/in/aswin_m |

<https://github.com/aswinhunter>

Education

Chennai Institute of technology, B.Tech, Artificial Intelligence and Data Science

2023 – 2027 - CGPA: 8.5/10

Internship

GARMENT TO PERSON CLOTH FITTING VTO Research, CIT Chennai (Offline)

Nov 2024 - Dec 2024

- Conducted literature review and comparative analysis of Virtual Try-On (VTO) systems.
- Studied and analyzed generative models including **GANs, cGANs, and Diffusion Models** for cloth fitting tasks.
- Performed data analysis and result validation for model evaluation.

Authored and submitted a structured review paper for publication

Projects

Dating Application (Real-Time Chat & Video Calling Platform)

Developed a full-stack dating application supporting real-time one-to-one chat and secure one-to-one video calls.

- Implemented secure login using Firebase Authentication and stored user/chat data in MongoDB.
- Built real-time group chat and one-to-one chat using Socket.IO connection.
- Random one-to-one video calling using WebRTC for peer-to-peer communication.
- Enabled user profiles, photo sharing, and similar to modern messaging platforms.

Tech Stack: Node.js, Express.js, Socket.IO, MongoDB, Firebase Auth, WebRTC, Selenium testing, Postman.

Agentic AI-based Interview Assessment

Designed and built a multi-agent AI system using LangGraph that autonomously conducts personalized mock interviews by orchestrating 5 specialized agents.

- Implemented a Resume Parser Agent using PyPDFLoader to extract a structured skill map from uploaded resumes, enabling every interview question to be tailored to the candidate's specific profile.
- Built a RAG-based Question Generator Agent using FAISS vector search and HuggingFace all-MiniLM-L6-v2 embeddings to retrieve semantically relevant questions from a curated question corpus.
- Developed an Answer Evaluator Agent that scores responses across technical accuracy, communication clarity, and a Follow-up Probe Agent that challenges vague or incomplete answers with targeted follow-up questions.
- Integrated OpenAI Whisper ASR for voice-enabled interviews and automated PDF performance report generation using reportlab - containing per-question scores, skill gap analysis, and a 4-week personalized study roadmap.

Tech Stack: Python, LangGraph, LangChain, Groq API (LLaMA 3), Gemini API, FAISS, HuggingFace Transformers, Streamlit.

Android application (agriculture app)

Built a smart agriculture Android application using Flutter with integrated weather forecasting and plant disease prediction feature.

- Weather prediction - Implemented weather forecasting using location/search with OpenWeatherMap API.
- Community - Built a community feature using Firebase Firestore for users to post and share insights.
- Mandi price - Automated price scraping from government websites using **Selenium** and **GitHub Actions**.
- Disease prediction - Integrated offline plant disease prediction using a **TensorFlow Lite model**.

Tools : Android studio, Selenium testing , web scraping, API, Flutter, Firebase, Figma.

Coding Platforms :

LeetCode : Max rating 1550, Top 49%, Solved 330 +Problems, Achieved highest rank 8,100.

CodeChef : Max rating 1250, 1 stars. || CodeForces : Max rating 1150.

Skills

Languages: Python, java, mysql, selenium, media pipe, opencv, pandas, tensorflow.

AI / ML: LLMs, Agentic AI, RAG, LangGraph, LangChain, FAISS, HuggingFace, Groq, Gemini, Prompt Engineering

DevOps: Git, GitHub, Docker, Kubernetes, Ansible. **Others:** Firebase, MongoDB, ServiceNow Dev Tool, WebRTC, Socket.IO

